



**Cambridge Instruments  
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · MOTOR-CUM-HAND OPERATED

# CIE/777 HM

Motor-cum-hand driven insulation tester — mains motor for PI testing, hand-crank backup for field use. Available in single-range configurations from 500V to 10,000V.

WOODEN BODY

## Motor-cum-Hand Driven Insulation Tester

Combines mains-motor drive with a collapsible hand-crank backup in one instrument — the same sustained-voltage PI and dielectric absorption capability as the CIE/777, plus a centrifugal-governed hand crank for use where mains power is unavailable.

- **Motor + hand-crank backup** — works with or without mains
- **Sustained PI & DAR testing** — motor-driven mode
- **Collapsible generator handle** — specific to the HM variant
- **Centrifugal governor** — constant voltage in hand-crank mode
- **Guard terminal & infinity adjuster** — eliminates surface-leakage errors



- M Motor + Hand** Works with or without mains
- P PI & DAR** Sustained voltage testing
- V 500-10000V** Available ranges, specify at order

*Manufactured in India since 1963 — trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors.*

### 01 Full Specification Sheet

| PARAMETER                          | SPECIFICATION                                                                            |
|------------------------------------|------------------------------------------------------------------------------------------|
| Drive                              | Mains motor (230V, 50Hz A.C.) + collapsible hand-crank                                   |
| Body                               | Seasoned wooden cabinet, sunmica finish, detachable lid, handle, lock                    |
| Scale Length                       | 100 mm (approx.)                                                                         |
| Accuracy                           | As per IS: 2992-1987                                                                     |
| High Voltage Test                  | Min. 2kV or (rated V + 1)kV A.C. r.m.s., 1 min                                           |
| Insulation Resistance Test         | Better than 50 MΩ at 500V D.C.                                                           |
| Guard Terminal & Infinity Adjuster | Guard ring & guard terminal eliminate surface-leakage errors; infinity adjuster provided |
| Response Time                      | 3 seconds                                                                                |
| Range Configuration                | Multiple range options available — specify at order                                      |
| Dimensions (top cover)             | 14 × 10 × 10 inch (approx.)                                                              |
| Weight (top cover)                 | 12 kg (approx.) — vs 9 kg for plain CIE/777                                              |
| Warranty                           | 12 months against manufacturing defects                                                  |

*Custom ranges and configurations can also be built against specific requirements.*

### 02 Typical Applications

- Field testing where mains supply is uncertain
- Sustained PI & DAR testing (motor-driven mode)
- Remote substation and railway traction commissioning
- HV transformer and cable insulation assessment

### 03 What's in the Box

- ✓ Mains cord, 2 metres
- ✓ 1 pair of H.V. testing leads

### 04 Optional Accessories

- + Rexine carrying case
- + 15-minute timer facility
- + Additional/special H.V. leads, 2m
- + Dual voltage & range option
- + Single voltage, dual range option

#### OFFICE

55, Ezra Street, 4th Floor, Yamuna Bhawan,  
Kolkata 700001, West Bengal, India

#### WORKS

23/2, Maulana Abul Kalam Azad  
Road, Howrah 71101, West  
Bengal, India

#### CONTACT

+91 98300 91260  
+91 98300 92117  
cieinstruments@gmail.com

#### ONLINE

cieinstruments.in  
Instruments for accuracy and  
reliability since 1963.